

Ultracold Fermions in an Ultrastable Optical Superlattice

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Upgrading the in-plane superlattice setup

At the start of my thesis, the optical setups for the $x1064$ lattice and the $x532$ lattice were already in place. The $x1064$ lattice was used in many former projects [1–4] in the original configuration. For the operation of the $x1064$ lattice as part of the in-plane superlattice, we only modified the optical setup on the laser table. This was required for the control of the superlattice phase, which I will discuss in detail in [<sec:phase-setup>](#). The $x532$ lattice was put into operation again shortly before my thesis [5]. On the experimental table, the optical setup was unchanged compared to the initial purpose of studying a Fermi gas in an optical superlattice [6].

After already working with the in-plane superlattice for more than three years, we noticed that the lattice depths V_{x1064} and V_{x532} are strongly time-dependent. While the intensity regulation with the photodiode on the experimental table showed a constant power, the lattice depth at the position of the atoms was dropping significantly in a few 100 ms. If the power is constant, the lattice beams could either change their positions or shapes to reduce the lattice depth at the position of the atoms. With the in-situ lattice modulation spectroscopy (see [<ch:mod>](#)), we could quantify the change of the lattice depth $V(x, y)$. In the $x1064$ lattice, we saw an increase of the Gaussian waist w_{x1064} that matched the decreasing lattice depth. We could confirm this behavior by imaging the forward-propagating lattice beam with a camera. For the $x532$ lattice, the measured reduction of the lattice depth was even stronger. Based on these observations, we concluded that the optical setups of the $x1064$ lattice and the $x532$ lattice are subject to severe thermal lensing. Initially, we made different attempts to compensate the changes of the lattice depths. However, due to the severity of the thermal lensing, we had to completely rebuild the optical setups of the two lattices on the experimental table.

In this chapter, I will introduce a theoretical description of thermal lensing to justify the choices of the optical materials that we used for the upgrade of the optical setup. As a reference for the upgrades and the other chapters in this thesis, I will show the complete optical setup of the $x1064$ lattice and the $x532$ lattice on the experimental table. This is also relevant for the alignment of the lattices in [<sec:mod-align>](#) and the control of the superlattice phase in [<sec:phase-setup>](#). At the end of the chapter, I will present the measured lattice

depths $V_{x1064}(\tau)$ and $V_{x532}(\tau)$, and compare them to the initial measurements to highlight the improvement of the optical setups.

3.1 Thermal lensing

In high-power laser applications, thermal lensing is a common issue [7]. The absorption of light in lenses, windows, and other optical elements induces a thermal gradient in the material that affects its geometrical and optical properties. Lasers used for welding or cutting can routinely achieve average powers of several kW, where elaborate techniques are required to compensate the effects of the thermal lensing. With a real-time focus-shift measurement, a closed-loop control system can be implemented to actively compensate the thermal lensing [8]. Another approach is to use specific materials to find a balance between thermal dispersion and surface deformation due to the absorption of light [9]. In the context of the experimental setup in this thesis, we are only dealing with intermediate optical powers up to a few watts. However, the collimated lattice beams have small diameters to realize the large beam waists at the atom position (see Section 3.2). Therefore, the beam intensity was sufficient to cause a thermal lensing that significantly affected the operation of the optical lattices. Compared to high-power applications, we could largely eliminate the thermal lensing by simply choosing the appropriate optical materials. Retrospectively, the thermal lensing was only enabled by a questionable material selection in the first place.

For the optical setup on the experimental table, we only consider thermal lensing due to the absorption of light in the bulk material and the dielectric coatings of the optical elements¹. The absorption of a laser beam induces a temperature gradient ΔT in the optical element. The dominant effects from the temperature gradient are a geometrical expansion of the material and a change of the refractive index. The expansion is proportional to the coefficient α , and the refractive index changes according to $\frac{dn}{dT}$. In total, we can quantify the optical effects of the temperature gradient ΔT with the coefficient

$$G = \alpha(n_0 - 1) + \frac{dn}{dT} \quad (3.1)$$

where n_0 is the refractive index of the optical material [10]. The *coefficient of thermal expansion* (CTE) α is positive for most optical materials², while the coefficient $\frac{dn}{dT}$ can either be positive or negative. For all optical materials with $\frac{dn}{dT} > 0$, the coefficient G is also positive. If the coefficient $\frac{dn}{dT}$ is negative, the value of G can range from weakly positive to weakly negative. Certain materials show an athermal behavior ($G \approx 0$) where the two contributions to the thermal lensing in Equation (3.1) compensate each other. If G is negative, the optical material can be used to compensate the thermal lensing of other components in composite optics [9].

The shape of the temperature gradient ΔT is not equal to the intensity profile of the laser beam. While the heating is proportional to the optical intensity, the temperature gradient

¹In industrial applications, scratches and other surface defects can also cause significant absorption [7].

²Corning Ultra-Low Expansion (ULE[®]) Glass [11] and SCHOTT ZERODUR[®] [12] are examples for glasses with a near-zero thermal expansion $\alpha = (0 \pm 10) \times 10^{-9} \text{ K}^{-1}$.

	UVFS	N-BK7	CaF ₂	N-SF5	N-SF6HT	N-SF11	N-BAF10	N-BALF4	TGG
$\rho/10^{-6} \frac{\text{W}}{\text{m}}$	6.15	4.08	-0.36	5.40	4.74	5.10	8.11	7.54	3.31
	6.60	4.69	-0.32	7.30	7.66	7.99	9.70	8.76	3.29
$a/\frac{\%}{10 \text{ mm}}$	< 0.001	0.12	< 0.1	0.24	0.12	0.08	0.24	0.28	0.24
	0.002	0.16	< 0.1	0.48	1.0	0.88	0.4	0.2	3.1

Table 3.1: Properties of the optical materials in the experimental setup. The thermo-optical coefficient ρ and the absorption coefficient a quantify the effective strength of the thermal lensing. Each coefficient has separate values for the wavelengths $\lambda = 1064 \text{ nm}$ (upper value) and $\lambda = 532 \text{ nm}$ (lower value).

- Add all references to data sheets etc. here...
- Add crystalline quartz (for waveplates)
- Highlight the rows to make it easier to see the different wavelengths?

ΔT distributes in the material based on the thermal conductivity k_T . The thermal lensing is therefore dynamic until a steady state is reached between the heating by the laser beam, the thermal conduction in the material and the dissipation of the heat from the surfaces. This matches our observation of the thermal lensing, where the lattice depth was changing rapidly in the first second, and significantly slower afterwards (see 3.3). For a qualitative statement on the suitability of different optical materials, we can use the thermo-optical ratio

$$\rho = G/k_T = \left(\alpha(n_0 - 1) + \frac{dn}{dT} \right) / k_T \quad (3.2)$$

that takes the thermal conductivity k_T into account as a mechanism to reduce the strength of the thermal lensing [10]. If the thermal conductivity is high, it prevents a large temperature gradient ΔT from building up in the optical elements. Nevertheless, we are not able to easily derive the actual shape of the thermal lens from the intensity profile of the laser beam. This would require a quantitative simulation of an elaborate heat model of the optical elements [9]. We only conclude that an optical element with a positive (negative) coefficient G forms a convex (concave) thermal lens in the center of the temperature gradient ΔT . The outer region of the temperature gradient has the opposite curvature, which results in a spherical aberration [7]. For the lattice beams, we are mainly interested in the focal shift in the center of the thermal lens, while the spherical aberrations are only a secondary effect. Since we are operating the lattice beams far away from the diffraction limit, small aberrations are negligible.

In Table 3.1, the relevant properties of all optical materials related to the $x1064$ -lattice setup and the $x532$ -lattice setup are compiled. The most common materials used for readily available optical elements are N-BK7 and UV-grade fused silica (UVFS). According to the thermo-optical ratio (3.2), they should experience a similar thermal-lensing strength. However, this does not take the initial heating due to the absorption of the laser beams into account. At the wavelengths 532 nm and 1064 nm, the typical absorption in UVFS is significantly lower than the absorption in N-BK7. If the absorption in the bulk material is the only heating source, UVFS is therefore

the ideal choice for most optical elements such as singlet lenses, windows and polarizing beam splitters (PBS).

Composite lenses such as achromatic doublets always use two materials with different dispersive properties to compensate chromatic aberrations [13]. The dispersion is quantified by the Abbe number V , where a small value corresponds to a strong dispersion and vice-versa. In an achromatic doublet, one lens is made from a *crown* glass which has a low refractive index and a weak dispersion, while the other lens is made from *flint* glass with a high refractive index and a strong dispersion. Both N-BK7 and UVFS are suitable crown glasses, although the former is more commonly used in achromatic doublets. Another typical crown glass that is also suitable for high-power applications is calcium fluoride (CaF_2). It has a very low absorption at 532 nm and 1064 nm, and the thermo-optical ratio is slightly negative. If the flint glass used for the other lens has a positive thermo-optical ratio, the achromatic doublet can be made athermal. In general, there is a wide range of crown and flint glasses available to construct achromatic doublets, and we had to check the thermal-lensing properties of the glasses individually for each material (see Table 3.1). This was essential for the replacement of the lens in the retro-propagating path. Regardless of the glasses used in an achromatic doublet, the lenses should be air-spaced for high-power applications. Cemented doublets can experience significant absorption in the material that is connecting the two lenses.

For polarization optics, the range of materials is limited compared to general optics such as lenses and windows. Waveplates require a birefringent material where the optical axis of the crystal is perpendicular to the propagation direction of the beam. In high-power applications, the most suitable material is crystalline quartz (SiO_2). Regardless of the orientation of the optical axis, it exhibits a very low absorption and ideal thermal properties [10]. In the orientation that is required for waveplates, the thermo-optical ratio almost vanishes completely. Analogous to achromatic doublets, compound waveplates are either air-spaced or optically contacted to avoid heating in the connecting material. Optical isolators use a Faraday medium in a magnetic field to rotate the polarization of the laser beam between two polarizing beam-splitters. The strength of the Faraday effect is characterized by the Verdet constant in units of $\frac{\text{rad}}{\text{T}\cdot\text{m}}$. The most commonly used material in optical isolators is terbium gallium garnet (TGG). It has a large Verdet constant and shows a low absorption at 1064 nm [14]. However, the absorption increases significantly towards shorter wavelengths [15]. At 532 nm, the absorption is more than $10 \times$ larger than the absorption at 1064 nm. Despite the large absorption, it is the default material used in readily available optical isolators at 532 nm. Other than selecting an optical isolator with a minimal crystal length, we can not reduce the thermal lensing at 532 nm. There is ongoing research on suitable optical materials with significantly lower absorption [16].

3.1.1 Simulating the focal shift

- Mention suppression of (thermal) lensing around the focus...?
- Interpret the scaling $\delta \propto f^2$?

Disregarding any higher-order aberrations, we can use a thin spherical lens to model the thermal lensing. Based on the observations in the experimental setup, we expect a thermal lens with

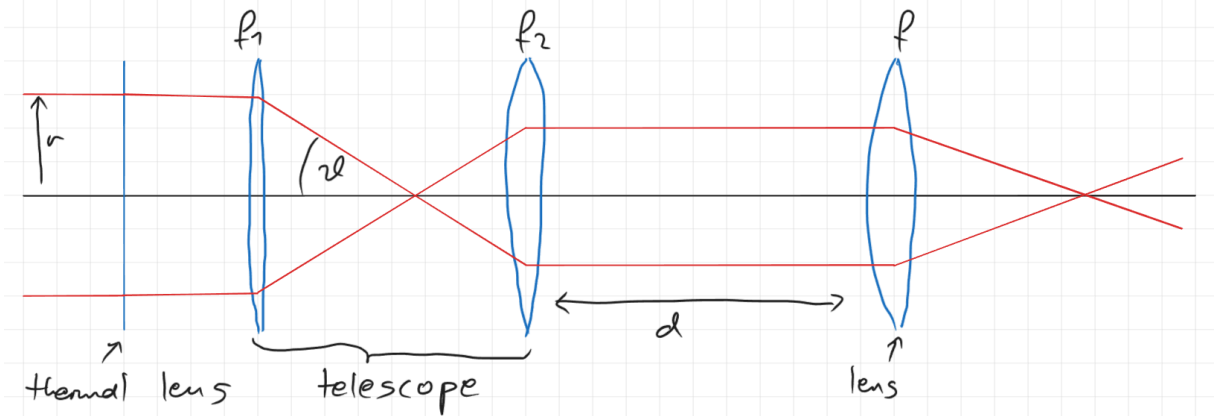


Figure 3.1: Optical setup for the simulation of the thermally-induced focal shift.

The beam is initially collimated with the radius $r = w_0$ and the angle $\vartheta = 0$. A thermal lens slightly focuses the beam before it is demagnified by the factor $\gamma = f_1/f_2$ in the telescope. After the beam shaping in the telescope, the beam is focused onto the position of the atoms by the lens with the focal length f . The dashed lines indicate the propagating beam without the thermal lens.

- Draw the rays without thermal lensing as dashed lines...

the focal length $f_{\text{thermal}} = \mathcal{O}(10 \text{ m})$. For a collimated lattice beam, this only amounts to a tiny change of the wavefront curvature. In order to see the focal shift at the position of the atoms, we need to focus the lattice beam with an additional lens of focal length f . If we neglect the propagation between the thermal lens and the additional lens, the focal shift amounts to

$$\delta = -pf^2 + \mathcal{O}(p^2) \quad (3.3)$$

where $p = 1/f_{\text{thermal}}$ is the optical power in units of m^{-1} . The negative sign corresponds to a focal shift towards the additional lens if the thermal lens is convex.

In Section 3.1 we concluded that the thermal lensing is amplified by the small beam diameters and the resulting high intensities. If we could use larger beams in the optical elements that are relevant for the thermal lensing, the optical power of the thermal lens will be reduced. To conserve the beam waist at the position of the atoms, we would then need a telescope that decreases the diameter of the collimated beams before they are focused onto the atoms by the lens with focal length f . Therefore, we include a telescope in the simulation to see how the focal shift changes compared to Equation (3.3). The total optical setup for the simulation is shown in Figure 3.1. The telescope is composed of two lenses with the focal lengths f_1 and f_2 at the distance $L = f_1 + f_2$, resulting in the demagnification $\gamma = f_1/f_2$. Behind the telescope, the beam will propagate by the distance d before it is focused onto the atoms. We use geometrical light rays instead of Gaussian beams to simplify the simulation. The effect of the divergence of the Gaussian beams on the focal shift is only a higher-order correction. In the ray transfer matrix formalism, we can describe the collimated beam with the vector

$$\mathbf{v}_0 = \begin{pmatrix} w_0 \\ 0 \end{pmatrix} \quad (3.4)$$

where the radius r corresponds to the waist w_0 of a Gaussian beam [17]. With the matrix equation

$$\mathbf{v}_1 = \underbrace{\begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix}}_{\text{focusing}} \cdot \underbrace{\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}}_{\text{propagation}} \cdot \underbrace{\begin{pmatrix} -1/\gamma & L \\ 0 & -\gamma \end{pmatrix}}_{\text{telescope}} \cdot \underbrace{\begin{pmatrix} 1 & 0 \\ -p & 1 \end{pmatrix}}_{\text{thermal lens}} \cdot \mathbf{v}_0 \quad (3.5)$$

we can compute the vector after the propagation through the optical setup in Figure 3.1. The position of the focus depends on the radius r_1 and the angle ϑ_1 of the vector $\mathbf{v}_1$. Relative to the expected distance f from the lens to the focus, we compute the focal shift

$$\delta = -\frac{r_1}{\vartheta_1} - f = -p\gamma^2 f^2 + \mathcal{O}(p^2) \quad (3.6)$$

that only differs by the factor γ^2 from the simple case in Equation (3.3). In the leading order, the focal shift depends neither on the initial beam waist w_0 nor on the distance d . With Gaussian beams, we would expect a small correction based on these two properties. The reason for the quadratic scaling of δ with the demagnification γ is hidden in the ray transfer matrix of the telescope in Equation (3.5). The radius is divided by the factor γ while the angle is multiplied by the factor γ . After the focusing lens, the reduced radius and the increased angle contribute equally to the focal shift $\delta \propto \gamma^2$. The quadratic scaling has a significant implication for the beam size in the thermal lens. Even if we use a larger beam diameter to reduce the thermal lensing, the focal shift is not actually reduced since we need to use a telescope with the demagnification γ after the thermal lens to conserve the beam waist at the position of the atoms. This assumes a linear scaling of the optical power p of the thermal lens with the maximum intensity $I_0 \propto w_0^{-2}$ in the center of the Gaussian beam. Nevertheless, it can be beneficial to use a larger beam for most of the optical setup to reduce the divergence due to the Gaussian nature of the lattice beams.

3.2 Experimental setup

For the upgrade of the optical setup to reduce the thermal lensing, we had to replace most optical elements that are used in transmission. However, we did not change the beam positions of the $x1064$ lattice and the $x532$ lattice. The overall geometry is therefore unchanged compared to the initial setup of the $x1064$ lattice [18,19] and the $x532$ lattice [5]. The upgraded version of the optical setup on the experimental table is shown in Figure 3.2. Unless noted otherwise, the transmissive optical elements are made of UVFS. In Subsection 3.2.1, I will introduce the specific replacements of the optical elements to reduce the thermal lensing.

The $x1064$ lattice uses a polarization-maintaining single mode fiber to guide the light from the optical table to the experimental table. Behind the collimator, the $x1064$ -lattice beam has a waist of $w_0 \approx 900 \mu\text{m}$. We use a 2 : 1 telescope to reduce the beam waist to $w_0 \approx 450 \mu\text{m}$. Even though the beam is technically collimated behind the telescope, the small waist results in a significant divergence of the beam up to the forward lens. This requires some additional

optical engineering to achieve the correct beam shape at the position of the atoms. Behind the telescope, the beam passes through an optical isolator³ to prevent the retro-propagating beam from realizing an additional standing-wave potential. With a $\lambda/2$ waveplate and a PBS, we rotate and clean the beam polarization which is essential for the interference of the lattice beams. The photodiode measures a fraction of the beam for the power regulation (see <sec:setup-lattices>). The last mirror in the isolated $x1064$ -lattice setup has always been used for the fine tuning of the lattice alignment. During the upgrade of the optical setup, we replaced the mechanical mirror mount with a piezo mirror mount⁴ to enable an automation of the $x1064$ -lattice alignment (see <ssec:mod-align-x1064>). Behind the alignment mirror, we use a relay lens with the focal length $f = 750$ mm to compensate the divergence of the beam behind the telescope. We determined the ideal position of the relay lens and the closest available focal length with a simulation of the propagation of a Gaussian beam in the optical setup. With the relay lens, we can continuously shift the focus around the atom position by changing the length of the telescope. This allows us to match the focus of the forward-propagating beam to the focus of the horizontal dipole trap. Without the relay lens, we can not move the focus to the atom position regardless of the telescope length.

The $x532$ lattice uses a polarization-maintaining photonic crystal fiber⁵ to guide the light from the optical table to the experimental table. Compared to a regular fiber, the photonic crystal fiber is more suitable for high-power applications due to the larger mode volume. The initial waist of the collimated beam is $w_0 \approx 700$ μm . For the optical isolation, we are using the Faraday medium from a commercially available optical isolator⁶ and two separate PBSs. The built-in PBSs of the optical isolator showed a significant contribution to the thermal lensing, most likely due to the optical material and a lack of optical contacting. The input PBS is rotated by 45° , which allows the output PBS to be parallel to the optical table. Behind the output PBS, we use the transmission of a backside-polished mirror for the power regulation. The subsequent telescopes prepare the correct beam shape for the $x532$ lattice. The first telescope uses cylindrical lenses for a vertical expansion of the beam by a factor 3. At the position of the atoms, this will translate to an aspect ratio of 1 : 3 between the vertical waist and the horizontal waist. In the second telescope, the overall beam size is reduced by a factor 2.5. The resulting horizontal beam waist is approximately 350 μm , which is required to achieve a waist of 120 μm at the position of the atoms. Similar to the $x1064$ lattice, the horizontal axis shows a strong divergence up to the forward lens. However, we decided against a relay lens since the intensity of the $x532$ lattice mainly depends on the position of the vertical focus. The details about the positioning of the separate foci are discussed in Subsection 3.3.2. Behind the second telescope are two mirrors in motorized mirror mounts⁷ for the alignment of the forward-propagating $x532$ -lattice beam onto the atoms. Due to the proximity of the mirror mounts compared to the distance to the forward lens, they are not suitable to shift the $x532$ -lattice beam perpendicular

³Conoptics M714 (TGG)

⁴Newport Agilis™ AG-M100N

⁵NKT Photonics LMA-PM-15

⁶Newport ISO-04-532-MP (TGG)

⁷Newport NewFocus 8821

to its optical axis. This is relevant to match the wavefronts of the $x532$ lattice and the $x1064$ lattice at the position of the atoms. To apply a horizontal and vertical shift to the $x532$ -lattice beam, we are therefore using two 10 mm thick windows. One window uses a static mirror mount, while the other window uses a piezo mirror mount⁸ to allow an automated control of the relative wavefront angle. Matching the wavefronts of the two lattices is essential for the phase-sensitive measurements presented in [<sec:phase-measure>](#).

The $x1064$ -lattice beam and the $x532$ -lattice beam are superimposed at the first dichroic mirror and focused onto the atoms by the forward lens⁹ with the focal length $f = 250$ mm. To avoid a normal incidence on the glass cell, the lattice beams are off-center with respect to the optical axis of the forward lens. The angle of incidence is approximately 2° in the xy plane and in the xz plane relative to the normal vector of the glass cell. This avoids an overlap of the lattice beams with the reflections off the glass cell, which would lead to additional interference patterns [18,19]. The distance from the forward lens to the atoms is determined by the horizontal dipole trap. With a vertical waist of $w_z \approx 12 \mu\text{m}$, it has a much shorter Rayleigh length than the optical lattices. We can therefore use the focus of the horizontal dipole trap to infer the position of the atoms relative to the forward-propagating lattice beams. With the retro-propagating lattice beams this is not possible, and we have to use an additional imaging system to find the correct retro-lens position. For an optimal overlap of the forward-propagating and retro-propagating lattice beams, the retro lens and the retro mirror have to be in a $4f$ -configuration. Otherwise, the divergence of the collimated lattice beams behind the retro lens causes focal shifts compared to the forward-propagating beams. For the fine tuning of the retro-lens position, we are using two imaging systems that measure the transmission of the lattice beams behind the dichroic mirrors around the glass cell. The beam-profiling imaging systems are also set up in a $4f$ -configuration using another lens with the focal length $f = 250$ mm and a camera at the virtual atom position. We can infer the center of the glass cell by looking for the outer surfaces of the glass cell. This measurement has an uncertainty of $\sigma \approx 1$ mm, which is negligible compared to the Rayleigh lengths of the lattice beams.

3.2.1 Replacement of optical elements

The optical setup of the $x1064$ lattice initially used two telescopes with focal-length ratios of $1.75 : 1$ and $1.25 : 1$. The total demagnification was therefore close to the ratio $2 : 1$ in the current setup. One telescope was located directly behind the collimator, and the other telescope was located just before the alignment mirror [18,19]. The second telescope might have been used for a similar purpose as the relay lens in the current optical setup. All four of the telescope lenses were made of N-BK7 and showed significant thermal lensing. We therefore removed both telescopes and replaced them with the single telescope and the relay lens shown in Figure 3.2. All three lenses are made of UVFS and do not show any measurable contribution to the thermal lensing. The only remaining optical element that causes thermal lensing is the optical isolator. Since the optical isolator already uses TGG as the Faraday medium, there is no readily available

⁸Newport Agilis™ AG-M100L

⁹LENS-Optics achromatic doublet ($\text{CaF}_2 + \text{N-BALF4}$)

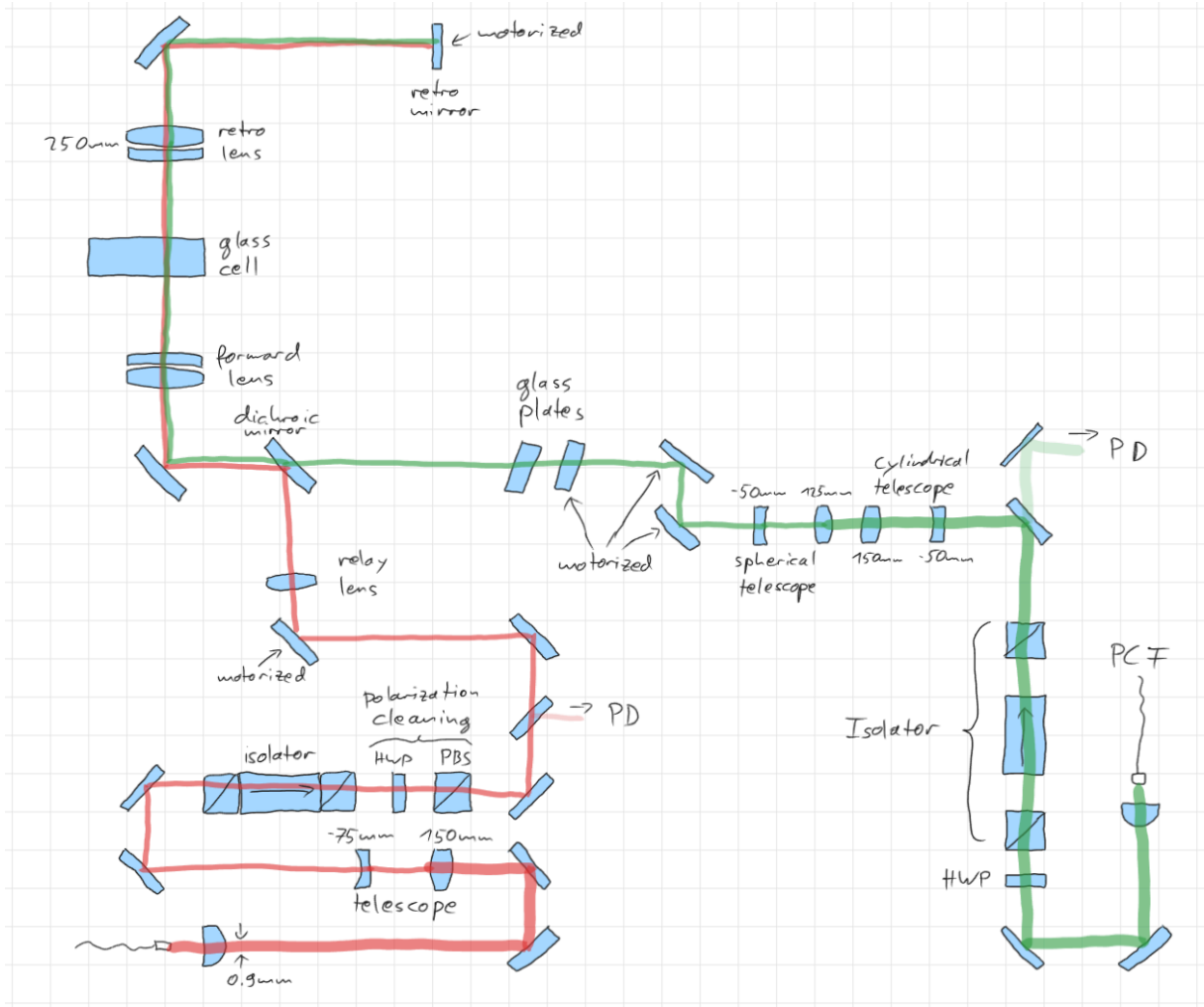


Figure 3.2: Experimental setup of the in-plane superlattice. The setup shows all relevant optics on the experimental table. To improve the visibility, the beam diameters are scaled up, and the distances are not to scale. Around the glass cell, the optical elements are shared with the imaging setup along the x axis (see [<ssec:setup-sequence-detect>](#)) and the horizontal dipole trap (see [<sec:setup-prepare-dipole>](#)). The imaging light with the wavelength $\lambda_{D2} = 766.7$ nm is transmitted by the dichroic mirrors around the glass cell. The forward lens and the retro lens have an anti-reflection coating designed for the wavelengths 532 nm, 766.7 nm and 1064 nm, just like the outer surfaces of the glass cell. The inner surfaces of the glass cell are uncoated, which results in a reflectivity of approximately 4 % at each inner surface. Between the relay lens and the following dichroic mirror, the horizontal dipole trap is overlapped with the $x1064$ lattice using a PBS.

- Indicate x-imaging path and horizontal dipole trap path?
- Mention optical paths for the forward and retro beam-profiling cameras?
- Add a legend for the lens properties?
- Add the PBS to overlap the $x1064$ lattice and the horizontal dipole trap?
- Assign numbers/labels to the mirrors and lenses?
- Reduce details in the caption?

replacement with a better Faraday medium. The only option for a reduction of the thermal lensing would have been a shorter optical isolator with a stronger magnetic field. Since the thermal lensing in the $x1064$ -lattice setup was already sufficiently small, we decided against pursuing any further improvements.

In the $x532$ -lattice setup, we initially replaced the collimated lens to increase the collimated beam size by the factor 2 in an attempt to reduce the thermal lensing. Correspondingly, we replaced the second lens of the spherical telescope by the equivalent achromatic doublet with the focal length $f = 50$ mm. We only understood the lack of improvement after the thermal-lensing simulation presented in Subsection 3.1.1. Nevertheless, we kept the larger collimated beam for the lower divergence up to the telescopes. After identifying the individual thermal-lensing contribution of the optical elements in the $x532$ -lattice setup, we replaced the four telescope lenses, the optical isolator and the PBS. The cylindrical telescope was composed of two N-BK7 lenses, which we replaced by their equivalent UVFS lenses. The spherical telescope used two cemented achromatic doublets made of N-BK7 + N-SF5 and N-BAF10 + N-SF6HT and respectively. Neither achromatic doublet was actually required given the small beam diameter, and we replaced both doublets with singlet lenses made of UVFS. The original optical isolator¹⁰ showed the greatest individual contribution to the thermal lensing of all optical elements in the $x532$ -lattice setup. We decided to replace it with a shorter optical isolator using a Faraday medium made of TGG¹¹. With the new optical isolator, the Faraday medium and each PBS contributed equally to the thermal lensing. We therefore removed the built-in PBSs and replaced them by optically-contacted PBSs made of UVFS. This leaves the Faraday medium as the last optical element that causes a small thermal lensing in the $x532$ -lattice setup. As presented in Subsection 3.3.2, we are able to find a beam configuration where the thermal lensing in the $x532$ lattice is effectively zero.

After upgrading the individual optical setups of the $x532$ lattice and the $x1064$ lattice, we could still observe a substantial thermal lensing contribution from the retro lens¹² in the shared optical path. The original retro lens was an achromatic triplet optimized for the wavelengths 532 nm and 1064 nm. While the triplet was air-spaced, the material composition was bad according to the thermal-lensing properties discussed in Section 3.1. Compared to the optical elements in the individual paths, we could observe a crosstalk between the two lattices since the beams are superimposed at the retro lens. This prompted us to replace the retro lens by the same model as the forward lens, where we could not observe any thermal lensing. Compared to N-BK7 and N-SF11, the optical materials CaF_2 and N-BALF4 have better thermal-lensing properties. Besides the focal shift of the lattice beams, the replacement of the retro lens also significantly improved the stability of the superlattice phase (see [<ssec:phase-sensors-stability>](#)).

¹⁰Conoptics M712A (Kigre M-18 glass)

¹¹We do not know the absorption coefficient at 532 nm and the thermal conductivity k_T of Kigre M-18 glass and can therefore not estimate the strength of the thermal lensing compared to TGG.

¹²CVI Melles Griot YAP-250.0-50.0 (N-BK7 + N-SF11)

3.3 Characterizing the optical lattices

- Mention the typical beam powers (at the atom position)?
- Already mention the superlattice phase φ here somewhere?

As already mentioned in the introduction of this chapter, we used the in-situ lattice modulation spectroscopy as well as beam-profiling cameras to characterize the thermal lensing. The modulation spectroscopy directly measures the spatial profile of the optical lattices, which reveals the lattice depth in the center, the Gaussian waist and the position. In terms of the measurement accuracy, this is the best characterization method available in the experimental setup. On the other hand, the beam-profiling cameras have a great time resolution and they can measure the beam profile around the atom position. This allows us to actually quantify the focal shift due to the thermal lensing, whereas the lattice modulation spectroscopy is restricted to the position of the atoms. Furthermore, the cameras can also resolve the vertical beam profile. The in-situ lattice modulation spectroscopy only shows the average horizontal lattice profiles in the occupied layers of the $z532$ lattice.

3.3.1 $x1064$ lattice

For the initial characterization of the thermal lensing in the $x1064$ -lattice setup, we used the in-situ lattice modulation spectroscopy. In the experimental sequence, we introduced a holding time τ just after the $x1064$ lattice is frozen at the lattice depth V_{x1064} (see [<sec:setup-sequence>](#)). For the characterization of the thermal lensing, we are neglecting the slow ramp up to $6 E_{\text{rec}}$ earlier in the sequence. The lattice-depth modulation begins right after the holding time τ and lasts 100 ms. In Figure 3.3a we can see an exponential decay of the lattice depth $V_{x1064} = 54 E_{\text{rec}}$ by 5 % up to a holding time of 1 s. During the rest of the holding time up to $\tau = 5$ s, the lattice depth decreases at the rate $0.25 \% \text{ s}^{-1}$. Overall, we observed an approximately linear strength of the thermal lensing. At $V_{x1064} = 45 E_{\text{rec}}$, the lattice depth only dropped by around 4 % in the first second. Running the in-situ lattice modulation spectroscopy down to lattice depths of $6 E_{\text{rec}}$ is not possible. We only could have measured the thermal lensing across the full range of the lattice depth with the beam-profiling cameras. The waist w_{x1064} in Figure 3.3b shows the complementary signal to the lattice depth with an exponential increase by approximately $20 \mu\text{m}$. From these two lattice parameters, we can conclude that the foci of the forward-propagating and retro-propagating beams were initially too close to their respective lenses. The thermal lensing then shifted the foci further towards their lenses to create a shallower but wider optical lattice at the position of the atoms. The other two lattice parameters did not show any relevant thermal lensing. At $\Delta y_0 < 1 \mu\text{m}$, the change of the lattice position is negligible compared to the waist w_{x1064} . The angle ϑ was constant at $-5.50^\circ \pm 0.21^\circ$ during the 5 s holding time, which matches the expected result in [<tab:mod-eval-results>](#).

For the initial thermal lensing, we do not have usable data available from the beam-profiling cameras. At the time, the imaging setup to measure the forward-propagating and retro-propagating lattice beams was not in a $4f$ -configuration. Due to the divergence of the collimated Gaussian beams, the cameras were therefore measuring the lattice beams at a significant distance from the atom position. Furthermore, we did not conduct measurements of the actual

focal shift caused by the thermal lensing. This requires a beam-profiling as a function of the holding time τ and the camera position x . We only developed the corresponding measurement routine after the upgrade of the $x1064$ -lattice setup.

After replacing all telescope lenses made of N-BK7 and the retro lens, we measured a focal shift $\delta < 1$ mm after 5 s at the lattice depth $V_{x1064} = 60 E_{\text{rec}}$. Relative to the Rayleigh length of $z_R \approx 60$ mm, this is a tiny shift that should result in a stability of the lattice depth better than 0.5 % if the foci are initially located at the atom position. After optimizing the focal position of the forward-propagating beam and the distance between the atoms and the retro lens with the cameras, we use the in-situ lattice modulation spectroscopy for the final characterization. In Figure 3.3a the lattice depth increases by less than 0.2 % in 5 s. Compared to the initial setup, we were able to reduce the variation of the lattice depth by a factor of approximately 30. Correspondingly, the waist w_{x1064} in Figure 3.3b does not show a drift in either direction. Besides being constant, the waist w_{x1064} is also significantly smaller in the final configuration. This allows us to achieve lattice depths beyond $60 E_{\text{rec}}$, while maintaining a waist that is still sufficiently large compared to the atom cloud. The stronger confinement along the y axis is not an issue since the $y1064$ lattice is always frozen when we are working with the in-plane superlattice. While we could significantly improve the stability of the lattice depth and the lattice waist, the position still drifts by up to $1 \mu\text{m}$ in 5 s. We can attribute this to the thermal cycle during the experimental sequence, which is mainly determined by the magnetic field coils (see <sec:setup-sequence> and <ssec:phase-sensors-stability>). In any case, the drift is negligible compared to the size of the atom cloud and the waist of the $x1064$ lattice.

3.3.2 $x532$ lattice

- Go into more detail about the “weird” horizontal focal shift? Also discuss the curvature then?

For the characterization of the $x532$ lattice, we used the in-situ lattice modulation spectroscopy in the superlattice potential (see <sec:mod-super>). The maximum $x532$ -lattice depth is not sufficient to run the measurement in the standalone lattice. In the superlattice potential, we can measure the combined depth of the $x1064$ lattice and the $x532$ lattice. Consequently, the $x1064$ -lattice depth should be constant during the measurement. Since we had not upgraded the optical setup yet, this required a compensation of the thermal lensing by applying the inverted signal in Figure 3.3a to the setpoint in the experimental sequence. With the standalone calibration of the $x1064$ lattice, we could achieve a stability of the lattice depth better than 0.6 %, which is already a $10 \times$ improvement compared to the initial configuration. In conjunction with the $x532$ lattice, we used the beam-profiling cameras to optimize the residual variation of the $x1064$ lattice. With an RGB camera¹³ we can capture $x1064$ -lattice beam and the $x532$ -lattice beam in the same image and extract the individual beam profiles during the evaluation. Due to the crosstalk of the lattice beams in the initial retro lens, we had to adjust the compensation of the $x1064$ lattice according to the $x532$ -lattice depth V_{x532} .

¹³Basler acA2040-35gc

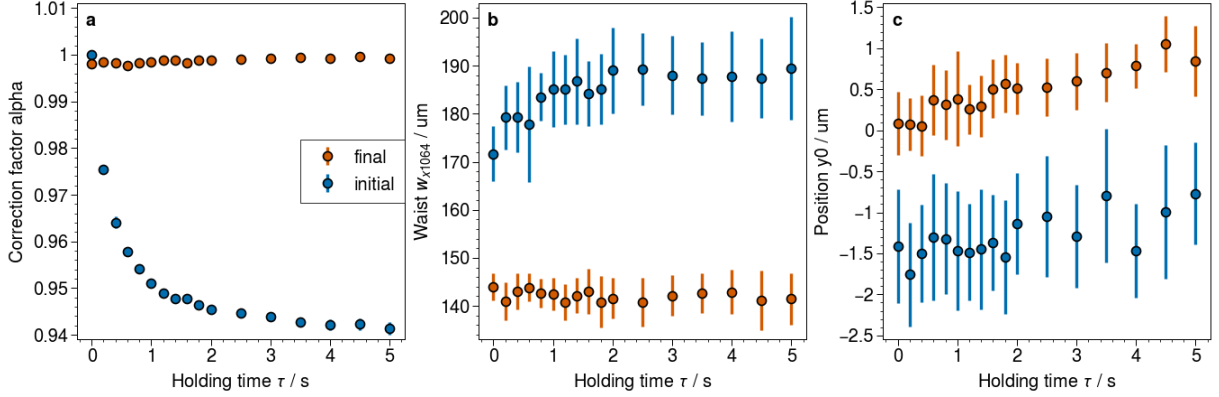


Figure 3.3: Improvement of the thermal lensing in the $x1064$ -lattice setup. The data shows the calibration of the lattice depth with the in-situ lattice modulation spectroscopy (see [<ch:mod>](#)). For the initial and final data, the lattice depth in the experimental sequence was set to $54 E_{\text{rec}}$ and $60 E_{\text{rec}}$ respectively. The modulation time was 100 ms, and we selected the modulation amplitude to obtain a good signal strength. The correction factor α quantifies the actual lattice depth relative to the setpoint in the experimental sequence. From the other lattice parameters (see [<fig:mod-eval-x1064-result>](#)), only the waist w_{x1064} and the position y_0 changed during the holding time, while the angle ϑ was constant in both setup configurations. The uncertainties of the lattice parameters are computed with the procedure introduced in [<ssec:mod-eval-error>](#).

- Move the **abc** indices outside of the axes? Just above the y-labels?

Before using the in-situ lattice modulation spectroscopy, we already measured the thermal lensing of the forward-propagating beam with the beam-profiling camera. However, as with the characterization of the $x1064$ lattice, the camera was not set up correctly to measure the beam profile at the position of the atoms. Nevertheless, we could already observe that the thermal lensing is even more severe for the $x532$ lattice compared to the $x1064$ lattice. The drop of the beam intensity was not linear as a function of the lattice depth V_{x532} anymore. Instead, the intensity appeared to saturate regardless of the setpoint V_{x532} . After a 1 s holding time at $V_{x532} = 24 E_{\text{rec}}$, the beam intensity was already reduced by 25 %, compared to a reduction by approximately 5 % at $V_{x532} \approx 14.5 E_{\text{rec}}$. The extrapolation of the beam intensity after a 3 s holding time suggests that the steady-state lattice depth is capped at $17 E_{\text{rec}}$. Instead of measuring the lattice depth as a function of the holding time τ , we therefore decided to run the in-situ lattice modulation spectroscopy as a function of V_{x532} after a 3 s holding time. This also allowed us to use a sufficiently long modulation time of 500 ms, which is the minimum recommended value to obtain a good signal with the superlattice modulation.

In Figure 3.4 we can see the thermal lensing in the initial optical setup. At the maximal lattice depth $V_{x532} = 24 E_{\text{rec}}$, the correction factor is $\alpha \approx 0.7$. From an interpolation of the data points in Figure 3.4a, we would expect the correction factor 0.85 at the lattice depth $V_{x532} \approx 18 E_{\text{rec}}$. This highlights the non-linearity of the thermal lensing in the $x532$ lattice. The horizontal waist in Figure 3.4b also changes as a function of V_{x532} , but not as expected from

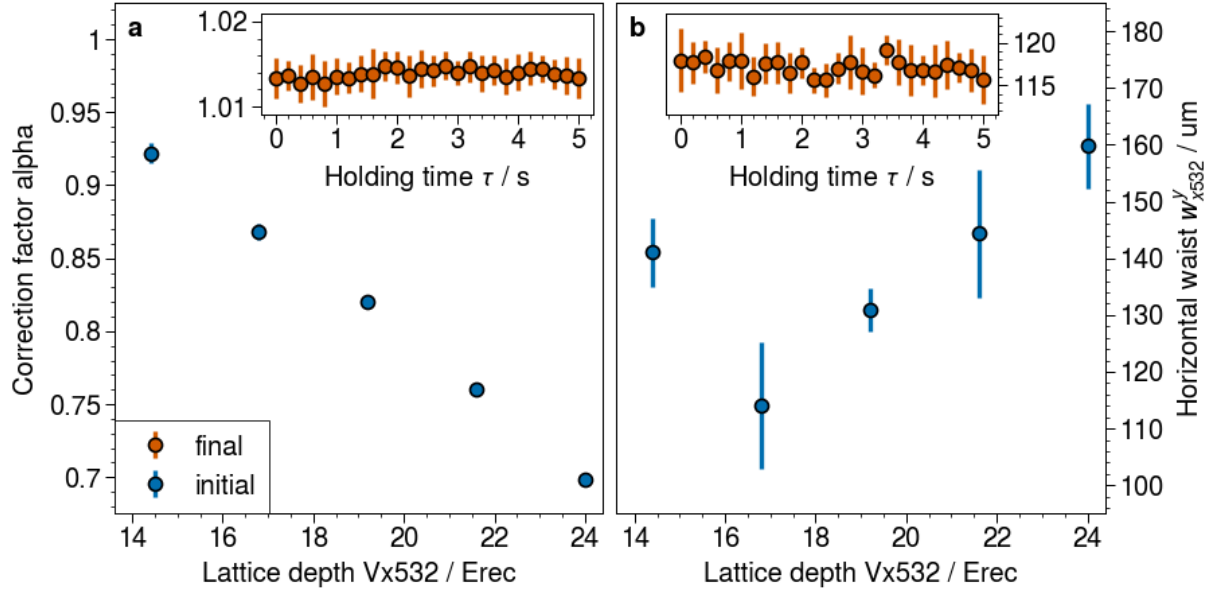


Figure 3.4: Improvement of the thermal lensing in the $x532$ -lattice setup. The initial data shows the results of the in-situ superlattice modulation spectroscopy at $V_{x1064} = 54 E_{\text{rec}}$ and $\varphi = \pi/4$ with a 500 ms modulation time (see [<sec:mod-super>](#)). We used a 3 s holding time approach the steady state of the thermal lensing. In **a**, we use the reference value $\alpha = 1.0$ from a measurement at $V_{x532} = 20 E_{\text{rec}}$ with a 100 ms modulation time after zero holding time. Due to the thermal lensing, this value is already slightly lower than the instantaneous lattice depth at $\tau = 0$. **b** only shows the horizontal waist w_{x532}^y since the in-situ lattice modulation spectroscopy is not able to resolve the vertical waist. Due to the ellipticity of the $x532$ -lattice beams, the two waists will evolve differently with the thermal lensing. The insets show the correction factor α and the horizontal waist w_{x532}^y in the final configuration as a function of the holding time τ with a 500 ms modulation time. The corresponding superlattice parameters are $V_{x1064} = 60 E_{\text{rec}}$, $V_{x532} = 18 E_{\text{rec}}$ and $\varphi = \pi/4$. The uncertainties of α and w_{x532}^y in the initial and final configuration are computed with the procedure introduced in [<ssec:mod-eval-error>](#).

- Add labels or legend for “initial” and “final”?
- Move the inset in **b** to the lower right corner? This would move it close to $115 \mu\text{m}$...

the correction factor α . Initially, the waist decreases to $114 \pm 12 \mu\text{m}$ before it increases to $160 \pm 8 \mu\text{m}$ again. We interpret this as shift of the horizontal foci of the forward-propagating and retro-propagating beams. The initial foci are too far away from their respective lenses, and the thermal lensing pulls the foci towards their lenses. At $V_{x532} \approx 17 E_{\text{rec}}$, the foci are closest to the atom position after a 3 s holding time. The drop of the lattice depth is a result of the combined changes in the horizontal direction and in the vertical direction. Based on the simulation of the thermal lensing in Subsection 3.1.1, the horizontal focus will be shifted approximately $9 \times$ further than the vertical focus. On the other hand, the vertical focus has a greater impact due to the shorter Rayleigh length. While we can not retrospectively quantify the actual focal shifts, we can estimate that they must be comparable to the Rayleigh lengths in order to reduce the

lattice depth $V_{x532} = 24 E_{\text{rec}}$ by 30 %. With the horizontal and vertical waists of $120 \mu\text{m}$ and $50 \mu\text{m}$, the corresponding Rayleigh lengths are approximately 85 mm and 15 mm.

In the final optical setup, neither the correction factor α nor the horizontal waist w_{x532}^y in Figure 3.4 show a change due to the thermal lensing. The variations of both parameters agree with the expected uncertainties of the in-situ superlattice modulation spectroscopy in [<tab:mod-super-result>](#). While the $x532$ lattice appears to be constant at the atom position, the thermal lensing is not actually zero. With the beam-profiling cameras, we can quantify the focal shifts around the atom position. At $V_{x532} = 18 E_{\text{rec}}$, the vertical focus is shifted by $\delta_y \approx 0.7 \text{ mm}$ and the horizontal focus is shifted by $\delta_z \approx 6 \text{ mm}$ after a 5 s holding time. If we look at the beam intensity, the focal shift amounts to $\delta \approx 0.9 \text{ mm}$. Realizing a constant lattice depth under these circumstances required an elaborate positioning of the foci. To minimize the change of the lattice depth, the vertical focus is located 2.5 mm away from the atom position while the horizontal focus is located 20 mm away from the atoms. We found this position empirically by looking at the beam intensity in the beam-profiling data. The maximum lattice depth is reduced by 3 % compared to the optimal focus position in exchange for a better stability of the $x532$ lattice.

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